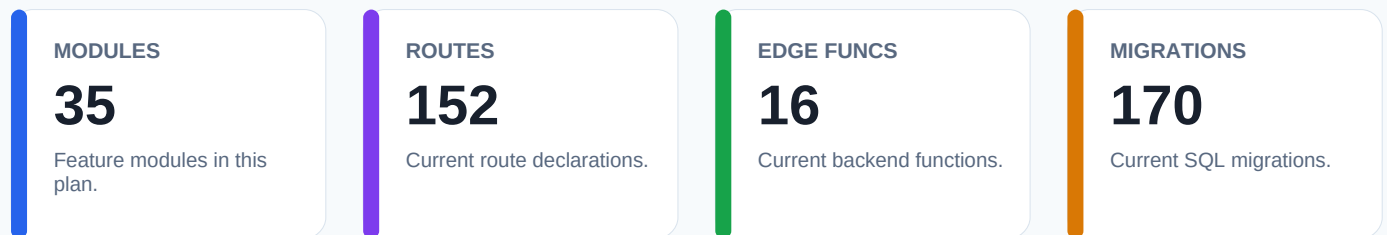


XO Platform

Vertical Slice Modular Monolith

Every module owns frontend, backend, shared contracts, and data boundaries

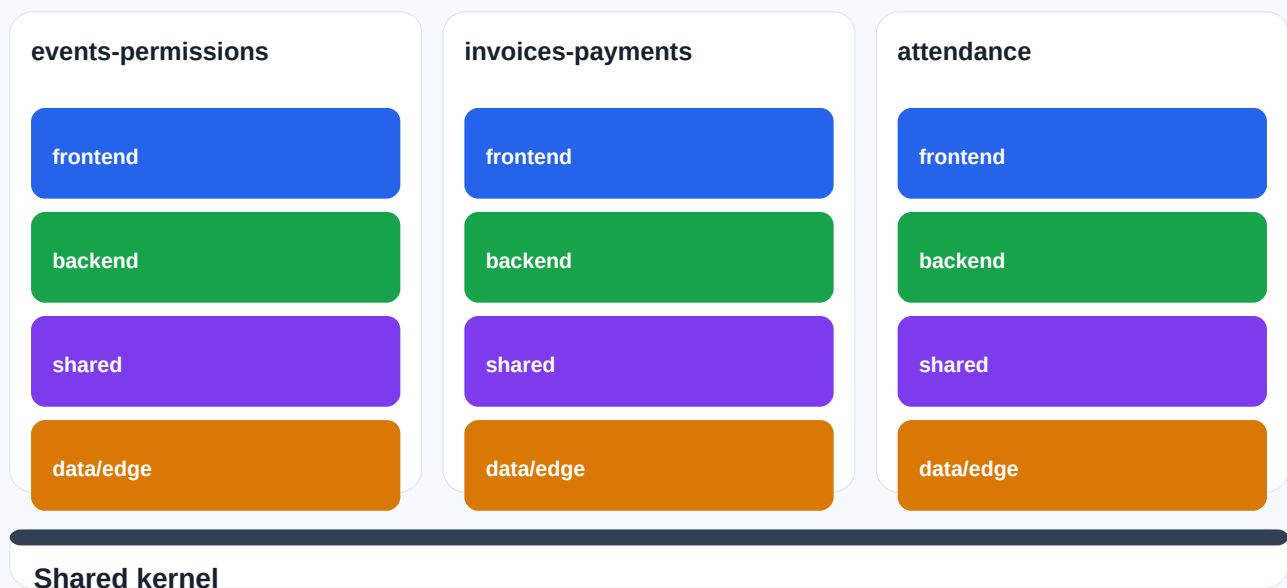


Meaning

A modular monolith is one deployable project, but its code is divided into strict modules. A vertical slice module contains the frontend screens, frontend hooks, backend handlers/functions, shared validation/types, and owned database objects for one feature.

One deployable system, many isolated modules

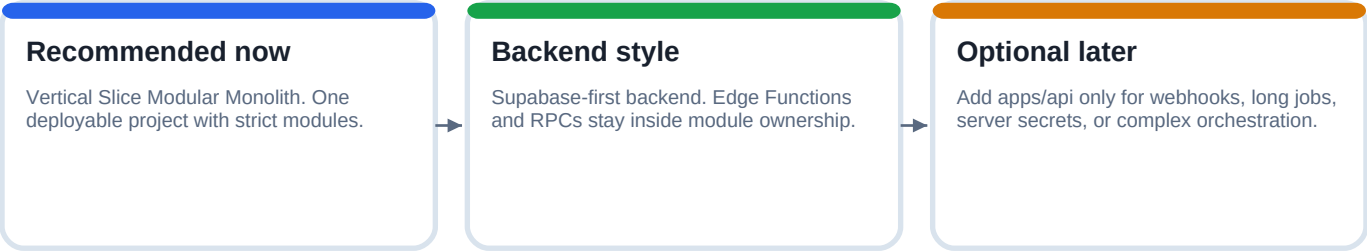
Each module owns frontend + backend + shared contracts. Modules communicate through public APIs, not deep imports.



Only generic UI, generated database types, permission contracts, validators, and pure utilities.

Recommended Structure Choice

For this project, the best balance is a Vertical Slice Modular Monolith with Supabase-first backend ownership.



Decision rule

Use modules for product features.
Use shared-kernel only for generic contracts, permissions, generated DB types, and utilities.
Use backend/supabase/modules/<module> for database, policies, and Edge Functions.
Use apps/api only when Supabase Edge Functions are no longer enough.

Question	Recommendation	Why it fits XO Platform
Should frontend and backend be split?	Yes, by module ownership, not by microservices.	The app has many business areas, but it still benefits from one repo and one coordinated deployment.
Should every module have frontend/backend/shared?	Yes.	When events, invoices, attendance, or staff changes, the full slice is easy to find and review.
Should every module use controller/service/repository?	Not now by default.	Supabase direct calls and Edge Functions are enough for most modules. Add controller/service/repository only inside apps/api later.
Where do Supabase functions live?	backend/supabase/modules/<module>/functions/<function>.	Each backend function gets an owner module and stops floating in a global folder.
Where do schemas and types live?	Inside the module shared folder, or shared-kernel if truly cross-cutting.	Validation contracts stay close to the feature that owns them.

Target Root Structure

This keeps the project deployable as one system while every feature remains separated by ownership.

Frontend App

```
XO-platform/  
  apps/  
    web/  
      public/  
      index.html  
      vite.config.ts  
      package.json  
      src/  
        app/  
          router/  
          providers/  
          layouts/  
          modules/  
            <module>/  
              frontend/  
                pages/  
                components/  
                hooks/  
                api-client/  
              shared/  
                schemas/  
                types.ts  
                constants.ts  
            index.ts  
          shared/  
            ui/  
            components/  
            hooks/  
            lib/  
          main.tsx  
          index.css
```

Backend, Shared Kernel, Ops

```
apps/  
  api/ # optional  
    src/modules/<module>/  
      backend/  
        routes.ts  
        controller.ts  
        service.ts  
        repository.ts  
        jobs.ts  
  
  backend/  
    supabase/  
      modules/  
        <module>/  
          migrations/  
          functions/  
          policies/  
          seed.sql  
          functions/_shared/  
          config.toml  
  
  packages/  
    shared-kernel/  
      src/  
        permissions/  
        generated-db/  
        common-types/  
        utils/  
  
  scripts/  
  docs/  
  tests/
```

One module owns the whole vertical slice

module = frontend + backend + shared contract + data ownership
example: events-permissions owns event pages, event hooks, event API calls,
event validation, event edge functions, event migrations, and event RLS tests.

Module Anatomy

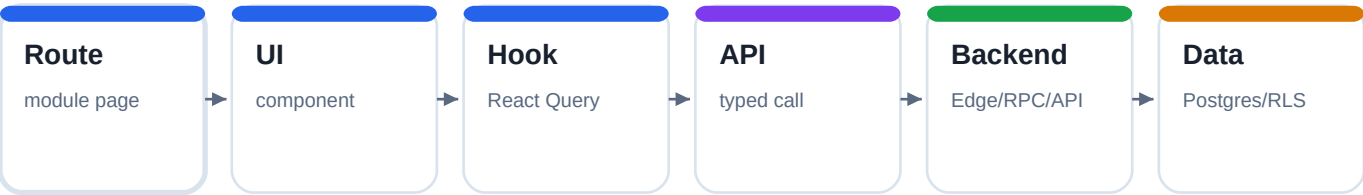
A feature module can be understood without jumping across the entire project.

```
modules/events-permissions/  
  frontend/  
    pages/  
      AdminEventsListPage.tsx  
      AdminEventDetailsPage.tsx  
      ParentEventsPage.tsx  
      TeacherEventsPage.tsx  
    components/  
      EventCard.tsx  
      EventFormFields.tsx  
      ParentPermissionCard.tsx  
    hooks/  
      useAdminEventsList.ts  
      useParentEventPermissions.ts  
      useTeacherEventsList.ts  
    api-client/  
      eventsClient.ts  
      permissionsClient.ts  
  backend/  
    edge-functions/  
      event-qr-issue/  
      permission-deadline-reminder/  
    db/  
      migrations/  
      policies/  
      seed.sql  
    services/  
      eventPermissionService.ts  
  shared/  
    schemas/  
      event.schema.ts  
      permission.schema.ts  
    types.ts  
    constants.ts
```

Part	What belongs here	What does not belong here
frontend/pages	Route-level screens and role variants.	Raw SQL/Supabase chains and complex business rules.
frontend/components	UI pieces used by this module.	Generic button/card/input primitives.
frontend/hooks	React Query hooks, form state, cache invalidation.	Backend secrets or direct service-role calls.
frontend/api-client	Typed client calls to Supabase, Edge Functions, or API modules.	UI rendering logic.
backend	Edge functions, API routes, domain services, database ownership.	React code.
shared	Zod schemas, DTOs, enums, constants used by frontend and backend.	Feature UI or server runtime code.
tests	Route smoke, service tests, RLS regression tests.	Manual checklist only.

How a Request Moves

This is the clean path inside one vertical slice. The page never reaches directly into unrelated modules.



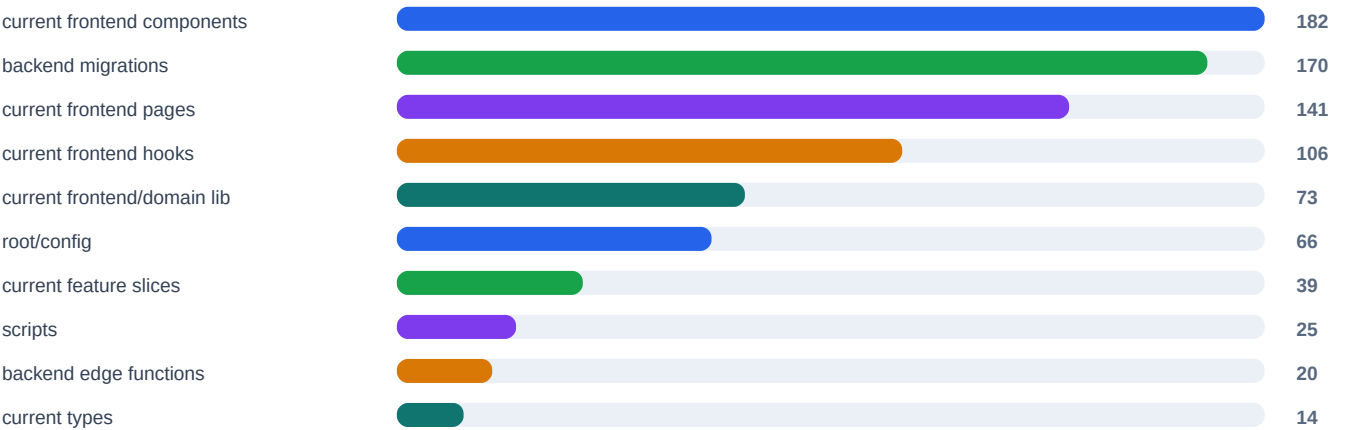
Step	Example in events-permissions	Rule
Route	/admin/events/:eventId	Route imports the module page from its public index.
Page	AdminEventDetailsPage	Composes UI and reads params only.
Hook	useAdminEventDetails	Owns loading/error/cache behavior.
Client	eventsClient.getEventDetails	Typed Supabase/RPC/Edge call boundary.
Backend	event permission service / edge function	Validates permissions and orchestration.
Database	events, event_permissions, event_attendees	RLS and migrations belong to backend ownership.

Allowed dependency direction

app shell -> modules -> shared-kernel
module frontend -> same module shared -> same module backend client
module backend -> same module shared -> shared-kernel/generated DB types
module A -> module B only through public index.ts or an explicit integration contract

Current Repo to Modular Monolith

The current project can move here gradually. The chart shows where code exists today.



Current folder	Target modular-monolith home	Action
src/pages/admin/AdminEvents*.tsx	apps/web/src/modules/events-permissions/frontend/pages	Move role pages into event module.
src/hooks/useAdminEventsList.ts	apps/web/src/modules/events-permissions/frontend/hooks	Move feature hook beside pages.
src/lib/event*.ts	apps/web/src/modules/events-permissions/frontend/api-client or shared	Split API client, schema, and pure helpers.
supabase/functions/event-qr-issue	backend/supabase/modules/events-permissions/functions/event-qr-issue	Move backend owner under module.
supabase/migrations/*event*.sql	backend/supabase/modules/events-permissions/migrations	Group migrations by owning module for readability.
src/lib/permissions/*	packages/shared-kernel/src/permissions	Permission engine is shared kernel.
src/components/ui/*	apps/web/src/shared/ui	Keep UI primitives outside modules.
src/components/layouts/*	apps/web/src/app/layouts	Layouts are app shell.

Backend Inside Each Module

Each module can use Supabase only, or Supabase plus a Node API module when the workflow needs it.

Backend option	Use for	Project example
Supabase direct client	Simple CRUD protected by RLS.	children profiles, inventory, meals, classes.
Supabase RPC	A database transaction or permission-sensitive query.	attendance summaries, finance rollups, set class lead.
Supabase Edge Function	Server-side workflow close to Supabase.	QR verification, import processing, AI assistant, dispatch, tenant export.
apps/api module	Long jobs, webhooks, external secrets, complex orchestration, or BFF endpoints.	future payment webhooks, external CRM sync, advanced reporting.

Backend module shape when apps/api is added

```
apps/api/src/modules/invoices-payments/  
  backend/  
    routes.ts      # HTTP routes / webhook routes  
    controller.ts  # request/response mapping  
    service.ts     # business workflow  
    repository.ts  # DB calls only  
    jobs.ts        # background jobs  
    validation.ts  # imports shared schemas  
    types.ts  
    index.ts
```

Shared Kernel

Shared kernel is intentionally small. It is the code that every module may import without creating messy dependencies.

```
packages/shared-kernel/src/  
  generated-db/  
    database.types.ts  
    tableNames.ts  
  permissions/  
    matrix.ts  
    can.ts  
    types.ts  
  common-types/  
    role.ts  
    locale.ts  
    money.ts  
    apiResult.ts  
  utils/  
    dates.ts  
    names.ts  
    phone.ts  
    assertNever.ts  
  index.ts
```

Allowed in shared kernel	Not allowed in shared kernel
Generated Supabase database types	Feature pages/components.
Permission matrix and can() helper	Feature-specific Supabase query chains.
Common enums and DTO primitives	Business workflows for one module.
Pure formatting helpers	React state that belongs to one UI.
Cross-cutting validation utilities	Large mixed helpers named utils for many domains.

Boundary test

If a file name contains a business noun like invoice, attendance, event, child, staff, media, QR, health, course, survey, or admission, it probably belongs inside that module, not shared kernel.

Module Summary

Every module owns a frontend area, backend area, shared contract, and data boundary.

Module	Roles	Frontend owns	Backend/data owns
Auth and Access	all users	routes for login/signup/unauthorized; Auth shell	Supabase Auth; data: users, staff_profiles, parents
XO Platform Admin	xo_super_admin	XO dashboard pages; nursery CRUD screens	tenant analytics queries; data: nurseries, payment_attempts, tenant_export_jobs
Tenant Settings and RBAC	xo admin, admin, manager	roles page; positions page	roles tables; data: roles, positions, features
Admin Dashboard	admin, manager	dashboard page; stats cards	dashboard queries; data: attendance_records, children, invoices
Nursery Setup Onboarding	admin, manager	onboarding steps; class setup UI	nursery settings; data: nursery_settings, classes, users
Children and Profiles	admin, parent	child list; child record	children table; data: children, child_guardians, child_avatars
Child Enrollment	admin, parent	child enrollment wizard; step panels	completion edge function; data: children, applications, child_documents
Parent Signup	public, parent	signup shell; signup steps	parent signup completion; data: parents, users, applications
Admissions and Applications	public, admin, parent	public inquiry form; inquiry kanban	inquiry/application tables; data: inquiries, waitlist, applications
Bulk Import	admin, operations	import wizard; column mapper	import jobs; data: import_jobs, children, parents
Attendance	admin, teacher, parent	admin attendance pages; teacher attendance page	attendance records; data: attendance_records, children, classes
Classes and Class Staff	admin, manager, teacher	classes list/detail; class staff editor	classes table; data: classes, class_staff, staff_profiles
Courses	admin, teacher, parent	course list; course form	course tables; data: courses, course_enrollments, children
Events and Permissions	admin, parent, teacher	event list/details/create/edit; permission cards	events; data: events, event_permissions, event_attendees
Invoices and Payments	admin, parent, finance	invoice CRUD; parent payment pages	invoice tables; data: invoices, invoice_line_items, payment_attempts
Financial Reports	admin, finance, xo admin	financial dashboard; revenue chart	finance views/RPCs; data: invoices, payment_attempts, packages
Staff and HR	admin, HR manager	staff directory; staff profile	staff profiles; data: staff_profiles, staff_documents, staff_schedules
Staff Onboarding	admin, HR manager	staff onboarding wizard; document uploads	staff onboarding completion; data: staff_profiles, staff_documents, users
Payroll and Payslips	admin, HR/finance, staff	payroll page; payslip create page	payroll records; data: staff_payroll_records, staff_profiles
Media Upload and Gallery	admin, teacher, parent	media upload; media approval	media table; data: media, children, classes
Daily Reports	teacher, parent, admin	daily report form; feeding/nap/mood/toilet sections	daily reports table; data: daily_reports, report_reactions, children
Milestones and Quarterly Reports	teacher, parent, admin	milestone list; milestone record page	milestones; data: milestones, quarterly_reports, children
Health and Safety	admin, parent, teacher	health profile; allergy/condition/medication dialogs	health records; data: child_health_records, child_health_documents, children
Chat and Messages	all roles	chat panel; chat widget	chat members; data: chat_conversations, chat_messages, chat_members
Broadcasts and Notifications	admin, teacher, parent	broadcast composer; broadcast history	notifications table; data: notifications, broadcast_messages, user_push_subscriptions
Surveys and Dynamic Forms	admin, parent	survey builder; form renderer	survey tables; data: surveys, survey_responses, notifications
Inventory	admin, manager	inventory page; stock list	inventory tables; data: inventory_items, inventory_movements
Meals and Nutrition	admin, teacher, parent	meal plans page; parent meals page	meal plan tables; data: meal_plans, child_dietary_preferences, daily_reports
Loyalty, Rewards, Packages	admin, parent, finance	loyalty dashboard; parent rewards	loyalty points; data: loyalty_points, loyalty_redemptions, packages
QR, Pickup, and Check-in	admin, teacher, parent	QR code cards; scanner page	QR token issuance; data: qr_tokens, pickup_verifications, attendance_records
Community and Content Library	admin, teacher, parent	community pages; content library pages	community content tables; data:

Module	Roles	Frontend owns	Backend/data owns
			community_posts, content_library_items
Personal and Teacher Reminders	admin, teacher, parent	reminders page; teacher reminders page	personal reminders; data: personal_reminders, notifications
AI Assistant and Help Center	all roles	AI assistant panel; help panel	AI assistant edge function; data: help_ai_preferences, ai_sessions
PWA, I18n, Theme	all roles	service worker bridge; install prompt	none by default; data: user_push_subscriptions, theme_preferences
Ops, Backup, Export	ops, admin	tenant export card; usage views if needed	tenant export edge function; data: tenant_export_jobs, audit_logs

Detailed Module Graphic

Frontend, backend, shared contract, data, and edge/API ownership for modules 1-2.

Auth and Access

Module: auth-access | Roles: all users | Priority: foundation

Frontend

routes for login/signup/unauthorized
Auth shell
session and route guard hooks

Backend

Supabase Auth
RLS helper functions
optional auth callback edge function

Shared

Role enum
auth session types
permission subject types

Data ownership

users, staff_profiles, parents

Edge/API ownership

none required now

XO Platform Admin

Module: platform-admin | Roles: xo_super_admin | Priority: core

Frontend

XO dashboard pages
nursery CRUD screens
platform analytics widgets

Backend

tenant analytics queries
nursery management tables
tenant export function

Shared

Nursery DTOs
platform analytics types

Data ownership

nurseries, payment_attempts, tenant_export_jobs

Edge/API ownership

tenant-export

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Detailed Module Graphic

Frontend, backend, shared contract, data, and edge/API ownership for modules 3-4.

Tenant Settings and RBAC

Module: tenant-settings-rbac | Roles: xo admin, admin, manager | Priority: foundation

Frontend

roles page
positions page
features page
permission gates

Backend

roles tables
role_features
position management policies

Shared

Feature keys
permission matrix
can() policy helper

Data ownership

roles, positions, features, role_features

Edge/API ownership

none required now

Admin Dashboard

Module: admin-dashboard | Roles: admin, manager | Priority: core

Frontend

dashboard page
stats cards
mini charts
activity feed

Backend

dashboard queries
attendance and finance rollups

Shared

dashboard view models
chart metric types

Data ownership

attendance_records, children, invoices, notifications

Edge/API ownership

none required now

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Detailed Module Graphic

Frontend, backend, shared contract, data, and edge/API ownership for modules 5-6.

Nursery Setup Onboarding

Module: nursery-onboarding | Roles: admin, manager | Priority: core

Frontend

onboarding steps
class setup UI
hours setup UI
invite teacher UI

Backend

nursery settings
class creation
self bootstrap policies

Shared

onboarding schema
working day constants

Data ownership

nursery_settings, classes, users

Edge/API ownership

none required now

Children and Profiles

Module: children-profiles | Roles: admin, parent | Priority: core

Frontend

child list
child record
parent child profile
avatar upload

Backend

children table
parent-child relationships
avatar storage policies

Shared

child profile types
age/name formatters

Data ownership

children, child_guardians, child_avatars

Edge/API ownership

none required now

Detailed Module Graphic

Frontend, backend, shared contract, data, and edge/API ownership for modules 7-8.

Child Enrollment

Module: child-enrollment | Roles: admin, parent | Priority: workflow

Frontend

child enrollment wizard
step panels
document upload UI

Backend

completion edge function
enrollment documents storage
application activation

Shared

child enrollment schema
file requirements
step validation

Data ownership

children, applications, child_documents

Edge/API ownership

child-enrollment-complete

Parent Signup

Module: parent-signup | Roles: public, parent | Priority: workflow

Frontend

signup shell
signup steps
file dropzone
success page

Backend

parent signup completion
parent account creation
uploaded documents

Shared

parent signup schema
contact validation

Data ownership

parents, users, applications

Edge/API ownership

parent-signup-complete

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Detailed Module Graphic

Frontend, backend, shared contract, data, and edge/API ownership for modules 9-10.

Admissions and Applications

Module: admissions-applications | Roles: public, admin, parent | Priority: workflow

Frontend

public inquiry form
inquiry kanban
waitlist
application review

Backend

inquiry/application tables
document verification
status transitions

Shared

application status enum
inquiry schema

Data ownership

inquiries, waitlist, applications, application_documents

Edge/API ownership

none required now

Bulk Import

Module: bulk-import | Roles: admin, operations | Priority: ops

Frontend

import wizard
column mapper
import review/progress pages

Backend

import jobs
AI mapper
background import processor

Shared

import row schema
validation result types

Data ownership

import_jobs, children, parents

Edge/API ownership

ai-import-mapper, process-import

Detailed Module Graphic

Frontend, backend, shared contract, data, and edge/API ownership for modules 11-12.

Attendance

Module: attendance | Roles: admin, teacher, parent | Priority: workflow

Frontend

admin attendance pages
teacher attendance page
parent attendance history

Backend

attendance records
attendance analytics queries
teacher toggle logic

Shared

attendance status enum
date range helpers

Data ownership

attendance_records, children, classes

Edge/API ownership

qr-verify when check-in uses QR

Classes and Class Staff

Module: classes | Roles: admin, manager, teacher | Priority: core

Frontend

classes list/detail
class staff editor
class announcements

Backend

classes table
class_staff policies
class lead RPC

Shared

class role types
class membership DTOs

Data ownership

classes, class_staff, staff_profiles

Edge/API ownership

none required now

Detailed Module Graphic

Frontend, backend, shared contract, data, and edge/API ownership for modules 13-14.

Courses

Module: courses | Roles: admin, teacher, parent | Priority: product

Frontend

course list
course form
teacher courses
parent courses

Backend

course tables
enrollment tables
teacher/course policies

Shared

course schema
course category enum

Data ownership

courses, course_enrollments, children

Edge/API ownership

none required now

Events and Permissions

Module: events-permissions | Roles: admin, parent, teacher | Priority: workflow

Frontend

event list/details/create/edit
permission cards
event QR cards

Backend

events
event permissions
event invoices
permission reminders

Shared

event schema
permission status enum
targeting types

Data ownership

events, event_permissions, event_attendees, event_invoices

Edge/API ownership

event-qr-issue, permission-deadline-reminder

Detailed Module Graphic

Frontend, backend, shared contract, data, and edge/API ownership for modules 15-16.

Invoices and Payments

Module: invoices-payments | Roles: admin, parent, finance | Priority: financial

Frontend

invoice CRUD
parent payment pages
payment attempt UI

Backend

invoice tables
payment attempts
Paymob initiation
reminders

Shared

invoice status enum
money formatter
payment DTOs

Data ownership

invoices, invoice_line_items, payment_attempts

Edge/API ownership

payment-reminders

Financial Reports

Module: financial-reports | Roles: admin, finance, xo admin | Priority: financial

Frontend

financial dashboard
revenue chart
payment charts
top parents

Backend

finance views/RPCs
invoice aggregation
payment analytics

Shared

finance metric types
date bucket helpers

Data ownership

invoices, payment_attempts, packages

Edge/API ownership

none required now

Detailed Module Graphic

Frontend, backend, shared contract, data, and edge/API ownership for modules 17-18.

Staff and HR

Module: staff-hr | Roles: admin, HR manager | Priority: ops

Frontend

staff directory
staff profile
documents
schedule editor

Backend

staff profiles
identity checks
HR manager policies

Shared

staff type enum
work schedule schema

Data ownership

staff_profiles, staff_documents, staff_schedules

Edge/API ownership

none required now

Staff Onboarding

Module: staff-onboarding | Roles: admin, HR manager | Priority: workflow

Frontend

staff onboarding wizard
document uploads
draft persistence

Backend

staff onboarding completion
profile mapping
document storage

Shared

staff onboarding schema
step error types

Data ownership

staff_profiles, staff_documents, users

Edge/API ownership

staff-onboarding-complete

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Detailed Module Graphic

Frontend, backend, shared contract, data, and edge/API ownership for modules 19-20.

Payroll and Payslips

Module: payroll-payslips | Roles: admin, HR/finance, staff | Priority: financial

Frontend

payroll page
payslip create page
staff payslip portal

Backend

payroll records
payslip visibility policies

Shared

payroll record type
currency formatters

Data ownership

staff_payroll_records, staff_profiles

Edge/API ownership

none required now

Media Upload and Gallery

Module: media-gallery | Roles: admin, teacher, parent | Priority: content

Frontend

media upload
media approval
parent gallery
teacher media list

Backend

media table
storage policies
privacy rules
approval workflow

Shared

media status enum
activity labels

Data ownership

media, children, classes

Edge/API ownership

none required now

Detailed Module Graphic

Frontend, backend, shared contract, data, and edge/API ownership for modules 21-22.

Daily Reports

Module: daily-reports | Roles: teacher, parent, admin | Priority: workflow

Frontend

daily report form
feeding/nap/mood/toilet sections
parent report detail

Backend

daily reports table
report reactions
media/report joins

Shared

daily report schema
care event enums

Data ownership

daily_reports, report_reactions, children

Edge/API ownership

none required now

Milestones and Quarterly Reports

Module: milestones-reports | Roles: teacher, parent, admin | Priority: learning

Frontend

milestone list
milestone record page
share cards
quarterly reports

Backend

milestones
quarterly reports
sharing rules

Shared

milestone types
quarterly report DTOs

Data ownership

milestones, quarterly_reports, children

Edge/API ownership

none required now

Detailed Module Graphic

Frontend, backend, shared contract, data, and edge/API ownership for modules 23-24.

Health and Safety

Module: health-safety | Roles: admin, parent, teacher | Priority: safety

Frontend

health profile
allergy/condition/medication dialogs
health alerts

Backend

health records
documents
parent update requests
alert computation

Shared

health schema
severity/status enums

Data ownership

child_health_records, child_health_documents, children

Edge/API ownership

none required now

Chat and Messages

Module: chat-messages | Roles: all roles | Priority: communication

Frontend

chat panel
chat widget
recipient picker
role inbox pages

Backend

chat members
messages
realtime publication
moderation fields

Shared

message DTOs
participant types

Data ownership

chat_conversations, chat_messages, chat_members

Edge/API ownership

none required now

Detailed Module Graphic

Frontend, backend, shared contract, data, and edge/API ownership for modules 25-26.

Broadcasts and Notifications

Module: broadcast-notifications | Roles: admin, teacher, parent | Priority: communication

Frontend

broadcast composer
broadcast history
notification drawer
push setup

Backend

notifications table
dispatch functions
push subscriptions
urgent broadcast

Shared

notification channel enum
broadcast schema

Data ownership

notifications, broadcast_messages, user_push_subscriptions

Edge/API ownership

email-dispatch, sms-dispatch, whatsapp-dispatch, emergency-broadcast

Surveys and Dynamic Forms

Module: surveys-forms | Roles: admin, parent | Priority: engagement

Frontend

survey builder
form renderer
survey analytics
parent survey page

Backend

survey tables
response storage
targeting notifications

Shared

survey schema
question type enum

Data ownership

surveys, survey_responses, notifications

Edge/API ownership

none required now

Detailed Module Graphic

Frontend, backend, shared contract, data, and edge/API ownership for modules 27-28.

Inventory

Module: inventory | Roles: admin, manager | Priority: ops

Frontend

inventory page
stock list
inventory filters

Backend

inventory tables
stock movement rules

Shared

inventory item types
stock status enum

Data ownership

inventory_items, inventory_movements

Edge/API ownership

none required now

Meals and Nutrition

Module: meals-nutrition | Roles: admin, teacher, parent | Priority: care

Frontend

meal plans page
parent meals page
teacher feeding section

Backend

meal plan tables
dietary preferences
feeding entries

Shared

meal schema
dietary preference types

Data ownership

meal_plans, child_dietary_preferences, daily_reports

Edge/API ownership

none required now

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Detailed Module Graphic

Frontend, backend, shared contract, data, and edge/API ownership for modules 29-30.

Loyalty, Rewards, Packages

Module: loyalty-rewards-packages | Roles: admin, parent, finance | Priority: engagement

Frontend

loyalty dashboard
parent rewards
points redemption
packages page

Backend

loyalty points
redemptions
packages/pricing tables

Shared

loyalty transaction types
package DTOs

Data ownership

loyalty_points, loyalty_redemptions, packages

Edge/API ownership

none required now

QR, Pickup, and Check-in

Module: qr-pickup-checkin | Roles: admin, teacher, parent | Priority: safety

Frontend

QR code cards
scanner page
pickup identity dialog
QR history

Backend

QR token issuance
QR verification
pickup snapshot
delegate pickup tokens

Shared

QR token types
scan parser
expiry labels

Data ownership

qr_tokens, pickup_verifications, attendance_records

Edge/API ownership

qr-token, qr-verify, event-qr-issue

Detailed Module Graphic

Frontend, backend, shared contract, data, and edge/API ownership for modules 31-32.

Community and Content Library

Module: community-content | Roles: admin, teacher, parent | Priority: content

Frontend

community pages
content library pages
content cards

Backend

community content tables
visibility policies

Shared

content type enum
audience types

Data ownership

community_posts, content_library_items

Edge/API ownership

none required now

Personal and Teacher Reminders

Module: personal-reminders | Roles: admin, teacher, parent | Priority: communication

Frontend

reminders page
teacher reminders page
alert host

Backend

personal reminders
repeat flags
notification scheduling

Shared

reminder recurrence types
reminder schema

Data ownership

personal_reminders, notifications

Edge/API ownership

none required now

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Detailed Module Graphic

Frontend, backend, shared contract, data, and edge/API ownership for modules 33-34.

AI Assistant and Help Center

Module: ai-help | Roles: all roles | Priority: support

Frontend

AI assistant panel
help panel
help preferences
route tracker

Backend

AI assistant edge function
AI safety/rate limit context
AI import mapper

Shared

AI tool types
help article metadata

Data ownership

help_ai_preferences, ai_sessions

Edge/API ownership

ai-assistant, ai-import-mapper

PWA, I18n, Theme

Module: pwa-i18n-theme | Roles: all roles | Priority: platform

Frontend

service worker bridge
install prompt
offline indicator
locales
theme switcher

Backend

none by default
push subscription storage for notification support

Shared

locale enum
theme preference type

Data ownership

user_push_subscriptions, theme_preferences

Edge/API ownership

none required now

Detailed Module Graphic

Frontend, backend, shared contract, data, and edge/API ownership for modules 35-35.

Ops, Backup, Export

Module: ops-backup-export | Roles: ops, admin | Priority: ops

Frontend

tenant export card
usage views if needed

Backend

tenant export edge function
integrity scripts
restore rehearsal
usage watch

Shared

export job status enum
ops result DTOs

Data ownership

tenant_export_jobs, audit_logs

Edge/API ownership

tenant-export

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Backend Function Structure Map

Every current Supabase Edge Function should have an owning module, shared request/response contract, backend implementation folder, and focused tests.

Function	Owner module	Recommended target folder
ai-assistant	ai-help	backend/supabase/modules/ai-help/functions/ai-assistant/
ai-import-mapper	bulk-import	backend/supabase/modules/bulk-import/functions/ai-import-mapper/
child-enrollment-complete	child-enrollment	backend/supabase/modules/child-enrollment/functions/child-enrollment-complete/
email-dispatch	broadcast-notifications	backend/supabase/modules/broadcast-notifications/functions/email-dispatch/
emergency-broadcast	broadcast-notifications	backend/supabase/modules/broadcast-notifications/functions/emergency-broadcast/
event-qr-issue	events-permissions	backend/supabase/modules/events-permissions/functions/event-qr-issue/
parent-signup-complete	parent-signup	backend/supabase/modules/parent-signup/functions/parent-signup-complete/
payment-reminders	invoices-payments	backend/supabase/modules/invoices-payments/functions/payment-reminders/
permission-deadline-reminder	events-permissions	backend/supabase/modules/events-permissions/functions/permission-deadline-reminder/
process-import	bulk-import	backend/supabase/modules/bulk-import/functions/process-import/
qr-token	qr-pickup-checkin	backend/supabase/modules/qr-pickup-checkin/functions/qr-token/
qr-verify	qr-pickup-checkin	backend/supabase/modules/qr-pickup-checkin/functions/qr-verify/
sms-dispatch	broadcast-notifications	backend/supabase/modules/broadcast-notifications/functions/sms-dispatch/
staff-onboarding-complete	staff-onboarding	backend/supabase/modules/staff-onboarding/functions/staff-onboarding-complete/
tenant-export	ops-backup-export	backend/supabase/modules/ops-backup-export/functions/tenant-export/
whatsapp-dispatch	broadcast-notifications	backend/supabase/modules/broadcast-notifications/functions/whatsapp-dispatch/

Function folder rule

```
backend/supabase/modules/<module>/functions/<function>/
  index.ts          # HTTP handler and orchestration entry
  contracts.ts      # imports shared schema/types when useful
  service.ts        # complex workflow only if function grows
  README.md         # purpose, env vars, input, output, owner
  __tests__/        # request validation, permissions, edge cases
```

Detailed Backend Function Graphic

Frontend, shared contract, backend files, and test structure for functions 1-2.

ai-assistant

Owner module: ai-help

Handles AI assistant requests, context, safety checks, and tool actions.

Frontend calls

AI assistant panel
Help panel
route tracker
preferences page

Shared contract

AI message DTOs
tool action types
safety result types

Backend files

index.ts
rate limit guard
context builder
tool dispatcher

Recommended target folder

backend/supabase/modules/ai-help/functions/ai-assistant/
index.ts
README.md
__tests__/

Tests to add

request validation
rate limit behavior
safe action approval flow

ai-import-mapper

Owner module: bulk-import

Maps spreadsheet columns into project import fields with AI assistance.

Frontend calls

Import wizard
column mapper
review page

Shared contract

import field schema
mapping confidence type
row validation result

Backend files

index.ts
AI mapping prompt
field normalization
mapping response parser

Recommended target folder

backend/supabase/modules/bulk-import/functions/ai-import-mapper/
index.ts
README.md
__tests__/

Tests to add

bad spreadsheet columns
unknown field mapping
fallback when AI fails

Detailed Backend Function Graphic

Frontend, shared contract, backend files, and test structure for functions 3-4.

child-enrollment-complete

Owner module: child-enrollment

Finalizes child enrollment after form submission and uploaded documents.

Frontend calls

Child enrollment wizard
document upload step
submit flow

Shared contract

child enrollment schema
document requirement types
completion payload

Backend files

index.ts
payload validator
child/application writer
document linker

Recommended target folder

backend/supabase/modules/child-enrollment/functions/child-enrollment-complete
index.ts
README.md
__tests__/

Tests to add

duplicate child guard
parent/admin access rules

email-dispatch

Owner module: broadcast-notifications

Sends email notifications for broadcasts, reminders, and operational messages.

Frontend calls

Broadcast composer
notification settings
history views

Shared contract

message channel enum
dispatch payload
delivery status type

Backend files

index.ts
provider adapter
template renderer
delivery logger

Recommended target folder

backend/supabase/modules/broadcast-notifications/functions/email-dispatch
index.ts
README.md
__tests__/

Tests to add

recipient
provider failure
delivery audit write

Detailed Backend Function Graphic

Frontend, shared contract, backend files, and test structure for functions 5-6.

emergency-broadcast

Owner module: broadcast-notifications

Creates urgent cross-channel broadcast messages for targeted users.

Frontend calls

Broadcast composer
urgent broadcast controls
history detail modal

Shared contract

urgent broadcast schema
target audience type
channel list

Backend files

index.ts
target resolver
notification writer
dispatch trigger

Recommended target folder

backend/supabase/modules/broadcast-notifications/functions/emergency-broadcast/
index.ts
README.md
__tests__/

Tests to add

target all parents
permission denial

event-qr-issue

Owner module: events-permissions

Issues QR tokens for event attendance or event check-in workflows.

Frontend calls

Event details page
event QR card
admin attendance section

Shared contract

event QR token type
expiry policy
issue request schema

Backend files

index.ts
event permission check
token generator
token storage

Recommended target folder

backend/supabase/modules/events-permissions/functions/event-qr-issue/
index.ts
README.md
__tests__/

Tests to add

expired event
wrong nursery
duplicate token handling

Detailed Backend Function Graphic

Frontend, shared contract, backend files, and test structure for functions 7-8.

parent-signup-complete

Owner module: parent-signup

Completes public parent signup and creates/links parent records.

Frontend calls

Parent signup page
signup review step
success page

Shared contract

parent signup schema
contact validation
signup completion payload

Backend files

index.ts
account linker
parent profile writer
application bootstrap

Recommended target folder

backend/supabase/modules/parent-signup/functions/parent-signup-complete
index.ts
README.md
__tests__/

Tests to add

duplicate email/mobile
invalid nursery
missing child data

payment-reminders

Owner module: invoices-payments

Finds due invoices and creates reminder notifications or dispatch jobs.

Frontend calls

Invoices page
payment record page
notification center

Shared contract

invoice status enum
reminder schedule type
payment reminder payload

Backend files

index.ts
due invoice query
notification creator
dispatch queue trigger

Recommended target folder

backend/supabase/modules/invoices-payments/functions/payment-reminders
index.ts
README.md
__tests__/

Tests to add

already paid invoice
overdue invoice
duplicate reminder prevention

Detailed Backend Function Graphic

Frontend, shared contract, backend files, and test structure for functions 9-10.

permission-deadline-reminder

Owner module: events-permissions

Reminds parents before event permission response deadlines.

Frontend calls

Parent permissions page
event details page
notification drawer

Shared contract

permission deadline type
event permission status
reminder payload

Backend files

index.ts
deadline scanner
pending response resolver
notification writer

Recommended target folder

backend/supabase/modules/events-permissions/functions/permission-deadline-reminder/
index.ts
README.md
__tests__/

Tests to add

deadline reminder /
already answered
multi-child event

process-import

Owner module: bulk-import

Processes import jobs after review, creating children, parents, and related rows.

Frontend calls

Import progress page
import review page
admin import wizard

Shared contract

import job status enum
validated row type
import error type

Backend files

index.ts
job loader
row processor
transaction writer
status updater

Recommended target folder

backend/supabase/modules/bulk-import/functions/process-import/
index.ts
README.md
__tests__/

Tests to add

partial failure
duplicate rows
rollback or error status

Detailed Backend Function Graphic

Frontend, shared contract, backend files, and test structure for functions 11-12.

qr-token

Owner module: qr-pickup-checkin

Issues child or delegate pickup QR tokens.

Frontend calls

Parent QR page
child QR card
delegate pickup QR card

Shared contract

QR token payload
expiry label type
delegate pickup schema

Backend files

index.ts
identity check
token generator
storage writer

Recommended target folder

backend/supabase/modules/qr-pickup-checkin/functions/qr-token/
index.ts
README.md
__tests__/

Tests to add

unauthorized parent
expired token policy
delegate metadata

qr-verify

Owner module: qr-pickup-checkin

Verifies scanned QR tokens and records pickup or attendance actions.

Frontend calls

Teacher scanner page
pickup confirm dialog
attendance UI

Shared contract

scan result type
QR verify request
pickup snapshot type

Backend files

index.ts
token verifier
attendance writer
pickup snapshot writer

Recommended target folder

backend/supabase/modules/qr-pickup-checkin/functions/qr-verify/
index.ts
README.md
__tests__/

Tests to add

expired token
wrong nursery
already used token

Detailed Backend Function Graphic

Frontend, shared contract, backend files, and test structure for functions 13-14.

sms-dispatch

Owner module: broadcast-notifications

Sends SMS notifications through the configured provider.

Frontend calls

Broadcast composer
notification preferences
delivery history

Shared contract

SMS dispatch payload
phone validation type
delivery status

Backend files

index.ts
phone normalizer
provider adapter
delivery logger

Recommended target folder

backend/supabase/modules/broadcast-notifications/functions/sms-dispatch
index.ts
README.md
__tests__/

Tests to add

build phone
provider error
audit logging

staff-onboarding-complete

Owner module: staff-onboarding

Completes staff onboarding and maps form payloads into staff profile records.

Frontend calls

Staff onboarding wizard
document uploads
admin staff page

Shared contract

staff onboarding schema
staff position type
completion payload

Backend files

index.ts
mapStaffProfile.ts
profile writer
document linker

Recommended target folder

backend/supabase/modules/staff-onboarding/functions/staff-onboarding-complete
index.ts
README.md
__tests__/

Tests to add

complete mode
new user mode
missing identity fields

Detailed Backend Function Graphic

Frontend, shared contract, backend files, and test structure for functions 15-16.

tenant-export

Owner module: ops-backup-export

Exports tenant-scoped data for backup, recovery, or operational review.

Frontend calls

Tenant export card
settings/export UI
ops dashboard if added

Shared contract

export job status
export manifest type
tenant scope type

Backend files

index.ts
tenant scope resolver
table exporter
archive writer

Recommended target folder

backend/supabase/modules/ops-backup-export/functions/tenant-export/
index.ts
README.md
__tests__/

Tests to add

cross-tenant denial
large tenant export
export manifest validation

whatsapp-dispatch

Owner module: broadcast-notifications

Sends WhatsApp notifications for broadcasts and reminders.

Frontend calls

Broadcast composer
channel checkboxes
delivery history

Shared contract

WhatsApp dispatch payload
channel enum
delivery result type

Backend files

index.ts
provider adapter
template renderer
delivery logger

Recommended target folder

backend/supabase/modules/broadcast-notifications/functions/whatsapp-dispatch/
index.ts
README.md
__tests__/

Tests to add

dispatch template
provider failure
delivery audit write

Migration Plan

Move one module at a time. Keep the app working after every wave.

Wave	Move	Result
1	Create app shell, modules folder, backend/supabase/modules, and shared-kernel package.	New structure exists without changing behavior.
2	Move generated DB types and permissions to shared-kernel.	Type drift and permission duplication start to shrink.
3	Move high-risk modules first: events, invoices, attendance, QR, onboarding.	Most change-heavy workflows become easier to maintain.
4	Move admin/parent/teacher pages into their owning modules.	Role pages for the same feature are close together.
5	Move Supabase edge functions and migrations under backend module owners.	Backend ownership becomes visible.
6	Add module-level tests: route smoke, service tests, RLS SQL tests.	Clean structure is protected by checks.
7	Only then consider apps/api modules for workflows Supabase cannot handle cleanly.	Avoid premature microservice complexity.

Definition of done for one migrated module

- [] Pages, components, hooks, API client, schemas, and types are inside the module.
- [] Backend functions/migrations/policies are owned by the module or documented if shared.
- [] Public imports go through module index.ts.
- [] No page contains raw business-heavy Supabase logic.
- [] npm run typecheck and npm run lint pass for changed files.
- [] At least one route smoke test or RLS test covers the module risk.

Appendix

Current repository facts used by this report.

Item	Detected value
Package	xo-platform
React	^19.2.4
Vite	^8.0.1
Supabase client	^2.100.0
React Router	^7.13.2
React Query	^5.95.2
Text/code files scanned	816
Approximate text/code lines	124,942
Route declarations	152
Supabase edge function index files	16
Supabase migration files	170

Current layer counts

Layer	Files
current frontend components	182
backend migrations	170
current frontend pages	141
current frontend hooks	106
current frontend/domain lib	73
root/config	66
current feature slices	39
scripts	25
backend edge functions	20
current types	14
docs	8
current providers	4
current router	3